

Prajwal Narendra Tidke

619-520-9475 | ptidke9@gmail.com | [LinkedIn](#) | [GitHub](#) | [Portfolio](#)

PROFESSIONAL SUMMARY

Data Engineer with 5+ years building batch and streaming pipelines across finance, manufacturing, and healthcare. Delivered high-scale lakehouse systems on AWS, Databricks, and Snowflake processing 1.2TB+ daily with PySpark, Airflow, and dbt - now building the production data backbone for RAG and agentic-AI systems.

EDUCATION

San Diego State University

MS, Big Data Analytics

Aug 2024 - May 2026

San Diego, CA

- **Achievements:** Academic Excellence Award for 4.0/4.0 CGPA

Indian Institute of Information Technology

BTech, Computer Science and Engineering

Aug 2016 - May 2020

Jabalpur, India

EXPERIENCE

San Diego State University - LINC Lab

AI Data Engineer

Feb 2025 - May 2026

San Diego, CA

- Built an ingestion pipeline parsing 1.5K+ multi-format clinical documents into a vector store via **Google LangExtract**, eliminating ~15 hrs/week of manual annotation.
- Engineered a **metadata-tagged chunking** and semantic-filtering retrieval layer over **ChromaDB** and **Azure OpenAI**, cutting query latency by **55%** (1.2s mean / 1.7s p95).
- Developed an automated evaluation harness (**RAGAS** + **LLM-as-a-judge**) gating answer quality at **0.96 faithfulness** across a 200-query test set, with **HIPAA-aligned guardrails** and a **5-stage state machine** orchestrating crisis responses for KALM.
- Ran a **multi-run ablation study** on a peer-knowledge augmentation layer, statistically validating and shipping the optimal configuration to production.

TresVista Analytics

Senior Data Engineer

Oct 2023 - Aug 2024

Pune, India

- Designed **ELT** pipelines on **Databricks Medallion Architecture** orchestrating portfolio analytics for a **\$550B+ AUM fund-**, improving processing reliability by **30%**.
- Orchestrated an **event-driven ingestion** (**AWS SQS/SNS to S3**) with optimized PySpark partitioning for a **1.2 TB/day** layer, and tuned high-cardinality tables via Delta Lake **Z-Ordering**, trimming median latency by **38%** and query execution times by **36%**.
- Integrated S&P Global and Visible Alpha feeds into financial models via a custom API connector and deployed **PyTest CI/CD data-quality gates**, reducing manual analysis by **45%** weekly while maintaining **100% SLA compliance** over 6 months.
- Mentored a team of 4 data analysts on engineering best practices and delivery standards, recognized with the **Best Mentor Award**.

LTIMindtree

Senior Data Engineer

Jul 2021 - Oct 2023

Pune, India

- Directed technical planning, code reviews, and Agile delivery across **5+ enterprise** manufacturing clients, achieving **100% SLA**.
- Led cloud migration of enterprise data infrastructure to **AWS Glue** and **Snowflake** with **SCD Type 2** tracking, saving ~\$650K (**65%**) in annual operating costs.

LTI

Data Engineer

Jul 2020 - Jul 2021

Pune, India

- Overhauled **batch ETL workflows** and tuned database performance through predicate pushdown and indexing, accelerating monthly regulatory reporting by **80%**.
- Implemented automated data-validation and profiling suites for **7+ downstream reporting pipelines**, detecting database anomalies and lowering production data errors by **63%**.

PROJECTS

Risk Pulse: Real-Time Portfolio Risk Intelligence | [GitHub Repo](#)

Jan 2026 - Apr 2026

- Architected a real-time **Apache Kafka** pipeline ingesting market tick data across **8 feeds**, computing rolling **VWAP/volatility** alerts and materializing to **Delta Lake** with an in-memory **DuckDB** engine serving **sub-150ms** queries to a live risk dashboard.
- Layered a **RAG service** (**ChromaDB** + **GPT-4.1**) over **dbt** mart snapshots, enabling natural-language portfolio-risk querying.

TrafficSensAI: Predictive Accident Analysis | [Live Link](#)

Aug 2024 - Dec 2024

- Constructed an end-to-end ML pipeline using **DBSCAN** and **Random Forest** on 487K US incident records, achieving an **86.3%** weighted F1 score for severity classification.
- Containerized the model with **Docker** and deployed an interactive hotspot map via the **ArcGIS** API, hosted on **Vercel**.

TECHNICAL SKILLS

- **Languages:** Python, Pandas, PySpark, Scala, SQL, Java
- **Data Engineering & Orchestration:** Airflow, Spark, Kafka, Flink, Hive, Hadoop, dbt, SCD (Type 1 & 2), ETL/ELT, Medallion Architecture, Dimensional Modeling, Data Quality, PostgreSQL, Oracle, DuckDB, Delta Lake
- **Cloud & DevOps:** AWS (S3, Glue, RDS, Athena, EMR, Lambda, SageMaker, CloudWatch), Azure, Databricks, Snowflake, Docker, Kubernetes, Terraform (IaC), BigQuery, CI/CD, FastAPI, Supabase
- **AI Systems:** LLMs, RAG, Agentic AI, Vector Database (ChromaDB), Azure OpenAI, Prompt Engineering, Claude Code, RAGAS

AWARDS & CERTIFICATION

- Business Unit of the Year (GoMX) - LTIMindtree (2022)

- DataCamp Certified Associate Data Engineer